Kinematic and Singularity Analysis Report with Explicit Matrix Operations

Automated Generation System

June 9, 2026

1 Mathematical Formulation

The closed-loop spatial kinematic constraint is given by:

$$T_1(q_1)T_2(q_2)T_3(q_3)T_4(q_4)T_5(q_5)T_6(q_6)T_7(q_7) = I \tag{1}$$

The velocity constraint yields $J_p\dot{\mathbf{q}}_p = -J_a\dot{q}_1$, where J_p is the 6×6 passive Jacobian matrix obtained by removing the column corresponding to the driving joint J_1.

2 Example Matrix Calculation at $q_1 = 0°$

To demonstrate the calculation process, we extract the system matrices at the initial driving configuration ($q_1 = 0$ rad). The solved full joint configuration (in radians) is:

$$\mathbf{q} = \begin{bmatrix} 0.0000 \\ 0.0166 \\ -0.0001 \\ -0.0331 \\ 0.0002 \\ -0.0166 \\ -0.0003 \end{bmatrix} \tag{2}$$

Based on this configuration, the full 6×7 Jacobian matrix J is evaluated as:

$$J = \begin{bmatrix} 0.000 & 0.000 & -0.000 & 2.500 & -0.041 & -2.500 & 0.000 \\ 0.000 & -1.000 & -0.000 & 3.000 & -0.000 & 1.999 & 0.000 \\ -0.000 & 0.000 & -3.000 & 0.041 & -0.500 & 0.041 & -2.000 \\ 0.000 & 0.000 & 0.000 & 0.017 & 0.000 & 0.017 & -0.000 \\ 0.000 & 0.000 & -1.000 & -0.000 & -1.000 & 0.000 & 1.000 \\ 1.000 & 1.000 & 0.000 & -1.000 & 0.000 & 1.000 & 0.000 \end{bmatrix} \tag{3}$$

By extracting the columns corresponding to the passive joints, we form the passive Jacobian $J_p \in R^{6\times 6}$:

$$J_p = \begin{bmatrix} 0.000 & -0.000 & 2.500 & -0.041 & -2.500 & 0.000 \\ -1.000 & -0.000 & 3.000 & -0.000 & 1.999 & 0.000 \\ 0.000 & -3.000 & 0.041 & -0.500 & 0.041 & -2.000 \\ 0.000 & 0.000 & 0.017 & 0.000 & 0.017 & -0.000 \\ 0.000 & -1.000 & -0.000 & -1.000 & 0.000 & 1.000 \\ 1.000 & 0.000 & -1.000 & 0.000 & 1.000 & 0.000 \end{bmatrix} \tag{4}$$

Applying Singular Value Decomposition (SVD) on $J_p = U\Sigma V^T$, the decomposition yields the left singular vectors U:

$$U = \begin{bmatrix} 0.666 & -0.064 & 0.639 & 0.001 & 0.380 & 0.001 \\ 0.671 & 0.019 & -0.738 & -0.004 & 0.068 & 0.007 \\ 0.039 & 0.987 & 0.059 & 0.142 & -0.001 & 0.000 \\ 0.003 & 0.000 & -0.005 & 0.000 & 0.007 & -1.000 \\ 0.006 & 0.141 & 0.014 & -0.990 & -0.006 & -0.000 \\ -0.323 & 0.027 & -0.208 & -0.007 & 0.923 & 0.007 \end{bmatrix} \tag{5}$$

The singular values matrix Σ (diagonal):

$$\Sigma = \begin{bmatrix} 4.177 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 3.670 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 3.354 & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 1.669 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.892 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \end{bmatrix} \tag{6}$$

And the transposed right singular vectors matrix V^T (representing joint couplings):

$$V^T = \begin{bmatrix} -0.238 & -0.029 & 0.958 & -0.013 & -0.154 & -0.017 \\ 0.002 & -0.846 & -0.025 & -0.172 & 0.072 & -0.500 \\ 0.158 & -0.057 & -0.121 & -0.021 & -0.978 & -0.031 \\ -0.002 & 0.338 & 0.002 & 0.551 & -0.008 & -0.763 \\ 0.958 & 0.011 & 0.258 & -0.010 & 0.123 & -0.005 \\ -0.014 & 0.408 & -0.008 & -0.816 & 0.005 & -0.408 \end{bmatrix} \tag{7}$$

3 Analysis Results & Curve Fitting

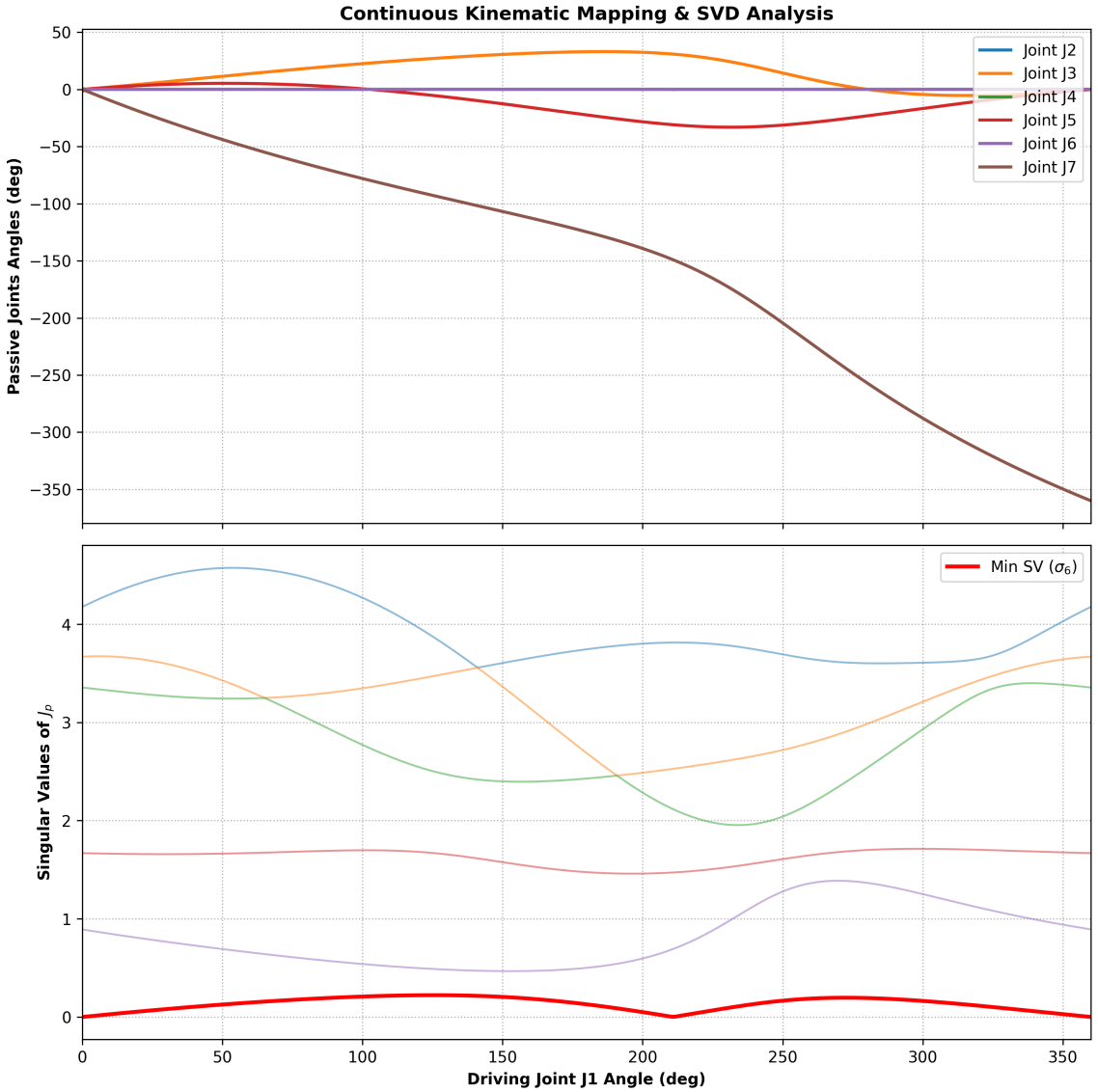


Figure 1: Kinematic mapping and singular value evolution driven by J_1.

The Fourier series expressions for the passive joints are:

- **Joint J2:**
  $Q_{J2}(x) = +0.003 + 0.004\cos(x) - 0.000\sin(x) + 0.005\cos(2x) + 0.001\sin(2x) + 0.005\cos(3x) - 0.001\sin(3x)$
- **Joint J3:**
  $Q_{J3}(x) = +14.740 - 17.160\cos(x) + 7.200\sin(x) + 2.093\cos(2x) + 3.424\sin(2x) + 0.617\cos(3x) - 0.949\sin(3x)$
- **Joint J4:**
  $Q_{J4}(x) = -0.005 - 0.008\cos(x) + 0.001\sin(x) - 0.010\cos(2x) - 0.001\sin(2x) - 0.009\cos(3x) + 0.001\sin(3x)$

- **Joint J5:**
  $Q_{J5}(x) = -11.728 + 11.633\cos(x) + 14.541\sin(x) + 0.433\cos(2x) - 1.922\sin(2x) - 0.460\cos(3x) + 0.224\sin(3x)$

- **Joint J6:**
  $Q_{J6}(x) = -0.003 - 0.004\cos(x) + 0.000\sin(x) - 0.005\cos(2x) - 0.001\sin(2x) - 0.005\cos(3x) + 0.001\sin(3x)$

- **Joint J7:**
  $Q_{J7}(x) = -153.532 - 28.794\cos(x) + 107.247\sin(x) + 1.660\cos(2x) + 62.635\sin(2x) + 1.077\cos(3x) + 37.015\sin(3x)$